Haresh Murugesan

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Education

The Ohio State University — Columbus, Ohio
Bachelor of Engineering in Computer Science, Specializing in AI, Minor in Business — *May 2027*

Experience

Technology Internship — May 2026 – August 2026
CAS — *Columbus, OH*
- Worked as an SW and AI Developer at CAS on Business and Agent-Based Applications for Product Stakeholders
- Utilized AWS services like S3 and DynamoDB to deploy applications, train AI Models, and process business data.
- Practiced Agentic and Trunk-Based Development practices while also assisting on C2 and C4 Architecture Designs

Undergraduate AI/ML Researcher — June 2026 – Present
Ohio State University — *Columbus, OH*
- Working on technical tasks in Digital Pathology using AI/ML methodologies

Innovation Analyst Internship — May 2025 – Present
Honda Research Institute Onramp Program — *Columbus, OH*
- Collaborated with 99P Labs in a data accelerator to identify unmet customer needs by applying lean startup and market research methodologies while working with leadership on 1000+ sample data models and visualizations

Founder and Interviewer — August 2023 – Present
Syntinuum: Endless Evolutions Podcast — *Dublin, OH*
- Founded and produced a tech podcast featuring CEOs and tech pioneers, growing the audience to over 8K YouTube views, 75 watch hours, and 50K Instagram views. 1000+ Likes and 50k+ Views on TikTok.
- Made a website using NextJS and unique loaders,animations,scrolls,etc. Found on SyntinuumPodcast.com

Projects

NYC Housing Crisis AI and ML Analysis Projects | *FastAPI, Pandas, Numpy* — Feb 2026
- Designed a 3D map utilizing custom CSS and WebGL layers to visualize building density and price disparities
- Implemented Unsupervised Machine Learning (K-Means Clusters) to segment NYC into 5 unique economic zones.
- Coded a "Market Force-Field" model using NumPy gradients to visualize housing price pressure as a vector field

AI and Algorithims (NLPs) Basics Projects | *Python, Numpy, TensorFlow* — July 2025
- Implemented an unbeatable Tic-Tac-Toe agent using the Minimax algorithm with alpha-beta pruning.
- Developed and trained a neural network using TensorFlow to recognize patterns in traffic data

TutorMatch AI Assisted Web App | *Python, Streamlit, HuggingFace AI* — June 2025
- Built a web application with a Streamlit frontend that leverages AI transformers from Hugging Face
- This application aims to improve student success and streamline the tutoring placement process in safe settings
- Used API keys, through frontend development and AI parsing through documents using PyPDF2

CarStomer : Car Value Analysis and Classification | *Python, Streamlit, Scikit-Learn, Pandas* — April 2025
- Launched a Streamlit and Pandas tool to predict and classify vehicle performance using a 6,000+ car dataset
- Optimized three Scikit-Learn models, with the Random Forest Regressor achieving an accuracy of 97.5 percent.

Technical Skills

Programming Languages: Java, Python, C, SQL, JavaScript, HTML/CSS, R, MATLAB, X86-64 Assembly
Coursework: Harvard's Intro to AI with Python, Data Structures and Algorithms, Systems Assembly, Discrete Structures, Propositional Logic, Calculus 1 and 2, Fundamentals of Engineering, Linear Algebra, Foundational Logic
Tools and Technology: React, TensorFlow, GitHub, NumPy, JUnit, Fast API, Avanti, Eclipse, Pandas, GDB Debugger

Organizations and Honors

First Place for Best use of External Data: ASA Datafest 2025
Top 3 Podcasts in Dublin, OH: *Adapt: Implementing Innovations Podcast*